11 · Case 8, part 1: the joint, A to H

The words of this track, as it says them. Bold brown is the examiner.

Exam case 8. Joints and arthritis.

Shannon is thirty, and has had a series of joint problems. She's had bursitis in her hip in the past, and now has pain mainly in her knee and hip joints, and her hands. She has recently been diagnosed with rheumatoid arthritis. She's concerned she's too young for arthritis, even though she's fairly active and has been an athlete most of her life. Part of her treatment plan is prednisone, a synthetic corticosteroid, similar to cortisol. Cortisol is part of the body's long-term stress response.

Case 8 is two tracks, and this one is the joint itself. She was clear in her session: the diagrams in the case studies will be in the exam, and she's promising you. This joint is definitely in, though it might be numbered instead of lettered.

Nine marks. Use the diagram to name the lettered structures of the joint, with the function of each.

Nine structures, a name and a job each. The hook, in letter order: muscle, enthesis, fluid, bursa, capsule, tendon, bone, cartilage, ligament.

Letter A, top left, is muscle. It contracts and acts on the bones, making the angle between them smaller or larger. Her answer deck letters the muscle as I, but the workbook figure letters it A. Go by the figure in front of you.

The other two A labels, top right and at the bottom, mark the enthesis: where the ligaments attach to the bone.

B is synovial fluid. It lubricates the joint.

C is a bursa. It reduces friction: it's a lubricated, fluid-filled sac between bone and soft tissue.

D is the joint capsule. It holds the bones together.

E is a tendon. It joins muscle to bone.

F is bone. It acts as a lever, and articulates at joints.

G is hyaline cartilage. It reduces friction, and absorbs shock.

H is a ligament. It holds bones together.

Her formative test drills the same diagram. B is synovial fluid, responsible for lubricating the joint. G is hyaline cartilage, and it is not responsible for secreting synovial fluid. The part arthritis mainly affects is G, the cartilage. And osteoarthritis occurs most often in the elderly, because of wear and tear.

Two marks. The knee and hip are both synovial joints. What are the benefits to the body of a synovial joint?

Synovial joints are freely movable. They're the most freely moving type of joint in the body. Without them, we'd have very limited movement.

Two marks. What kind of synovial joint are the hip and the knee? The shoulder and the elbow?

The shoulder and the hip are ball-and-socket joints: a ball moving in a socket. The knee and the elbow are hinge joints: they can only open and close, like a door.

Two marks. What is bursitis?

A bursa is a thin, lubricated, fluid-filled sac between a bone and the soft tissue around it, the tendons and muscles. It's a cushion to reduce friction. In bursitis, it becomes inflamed and swollen. Then it has the opposite effect, and it's painful.

And one she hasn't answered anywhere yet. Compare and contrast bursitis and arthritis.

Her answer isn't posted, so this is built from her two separate answers. What they share: both are inflammation, with pain and swelling around a joint, and both can limit movement. Where they differ. Bursitis is in the bursa, a fluid sac outside the joint, often from overuse, pressure or injury, and it usually settles. Arthritis is in the joint itself. In her words, arthritis wears away the articular cartilage and inflames the synovial membrane. And it's usually long-term. Shannon has had both: bursitis in her hip before, and rheumatoid arthritis now.

Once more. The joint.

The joint, letter by letter: muscle contracts to move the bones; the enthesis is where ligaments attach; synovial fluid lubricates; a bursa reduces friction; the joint capsule holds the bones together; a tendon joins muscle to bone; bone is a lever that articulates; hyaline cartilage reduces friction and absorbs shock; and a ligament holds bones together. Synovial joints are freely movable. The shoulder and hip are ball-and-socket; the knee and elbow are hinges, like a door. Bursitis is an inflamed, swollen bursa, the friction cushion, and it's painful.